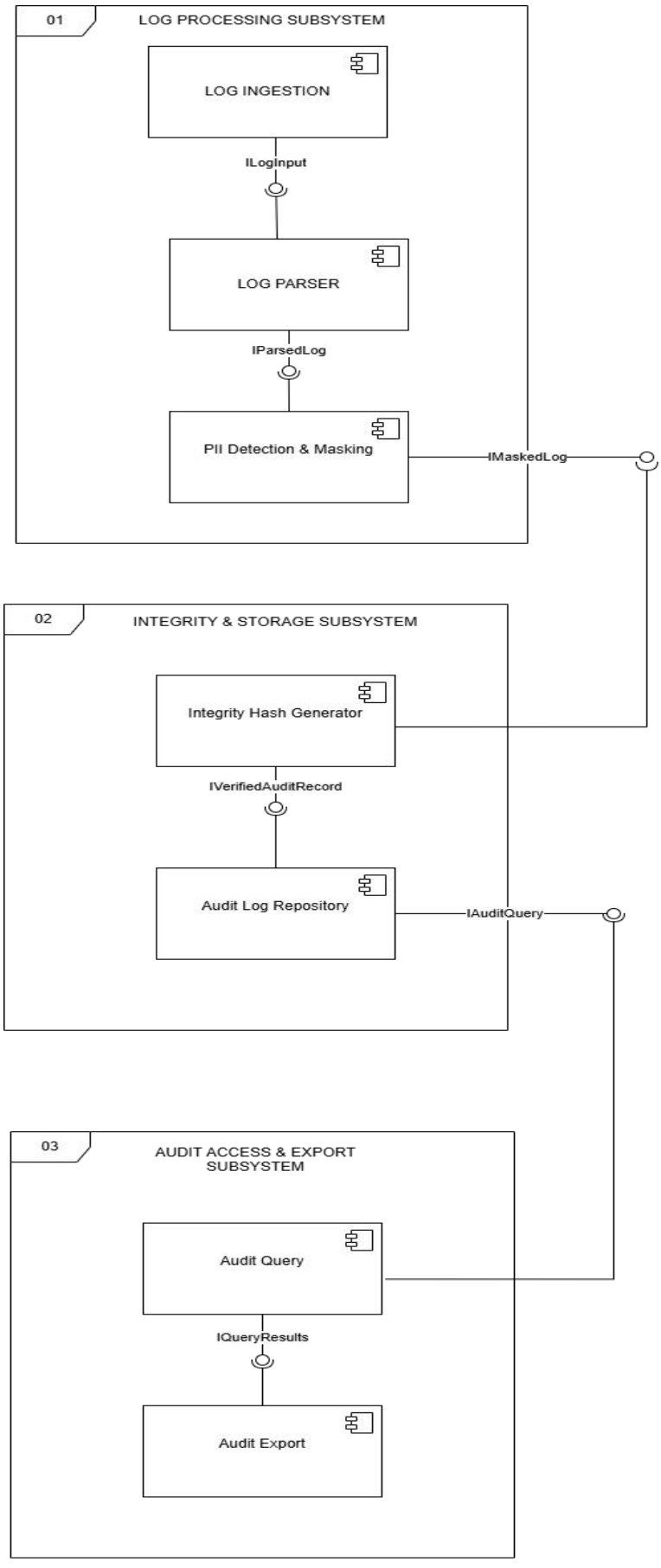
SOFTWARE ENGINEERING

NAME : B THRIVIKRAM
SRN : PES1UG24AM423
SECTION : H
LAB : 03


01
LOG PROCESSING SUBSYSTEM
LOG INGESTION
ILogInput
LOG PARSER
IParsedLog
PII Detection & Masking
IMaskedLog
02
INTEGRITY & STORAGE SUBSYSTEM
Integrity Hash Generator
IVerifiedAuditRecord
Audit Log Repository
IAuditQuery
03
AUDIT ACCESS & EXPORT SUBSYSTEM
Audit Query
IQueryResults
Audit Export

Architecture Selection: Layered Architecture

We chose the **Layered Architecture** for the **Centralized Audit Trail Compliance Engine**.

The first reason for selecting this architecture is **clear separation of concerns**. The system performs several different tasks, including ingesting enterprise application logs, parsing structured JSON logs, detecting and masking Personally Identifiable Information (PII), generating integrity hashes, storing audit records, querying historical logs, and generating audit exports. By separating these responsibilities into different components and subsystems, the system becomes easier to understand, maintain, test, and modify.

The second reason is that the system requires a **structured and secure data-processing flow**. In the proposed architecture, incoming logs first pass through the Log Processing Subsystem, where they are parsed and sensitive PII is detected and masked. The masked logs are then passed to the Integrity and Storage Subsystem, where a SHA-256 integrity hash is generated before the audit record is stored. Finally, the Audit Access and Export Subsystem allows authorized users to query audit records and generate audit exports. This clear flow ensures that each stage performs its specific responsibility.

A major **security advantage** of the selected architecture is that sensitive PII can be masked before the audit record is persisted in the Audit Log Repository. This helps prevent sensitive information, such as credit card numbers and Social Security Numbers, from being stored in cleartext. Additionally, generating a SHA-256 integrity hash helps verify the integrity of audit records and detect possible tampering.

A key **performance benefit** is that separating log processing, storage, and audit access into different subsystems allows each part of the system to be optimized independently. In particular, the Audit Log Repository and Audit Query components can be optimized for efficient searching and filtering of large volumes of historical audit records, while log ingestion and PII masking can operate independently of query operations.

Therefore, the **Layered Architecture** provides a clear, modular, secure, and maintainable solution for the Centralized Audit Trail Compliance Engine. It supports the required separation between log processing, integrity and storage, and audit access while addressing the important security and performance requirements of the system.